

MOHAMED MUFASSIR

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PROFESSIONAL SUMMARY

3rd-year Computer Systems Engineering undergraduate (SLIIT) specializing in embedded systems, robotics, and real-time control systems. Proven ability to design and build intelligent hardware-software systems — from AVR-programmed autonomous robots to IoT-connected vehicles with live video streaming. Supplemented by industry internship experience in software development. Passionate about robotics, industrial electronics, and building embedded solutions that operate reliably in the real world.

EDUCATION

BSc (Hons) in Computer Systems Engineering

Nov 2023 – Expected Nov 2027

Sri Lanka Institute of Information Technology (SLIIT) | GPA: Upper Second Class (3.0–3.5)

Relevant Modules: Real-Time Systems, Digital Logic Design, Control Theory, Computer Architecture, Data Structures & Algorithms, Computer Networks

EXPERIENCE

Software Engineering Intern — ZILLIT (Startup), Nittambuwa

Nov 2025 – Feb 2026

- Led sole development of a cross-platform Point of Sale (POS) desktop application using Electron.js and TypeScript, designed as part of a broader IoT ecosystem (desktop + mobile integration)
- Built the front-end of an enterprise-level financial management system using Vue.js, integrating with a Java Spring Boot backend
- Worked with React, TypeScript, Vue.js, Electron.js, PostgreSQL, and Docker in a fast-paced startup environment
- Collaborated on system architecture decisions for scalable, production-ready software

Freelance Web Developer — Self-Employed, Local & Online

2024 – Present

- Designed and developed business websites for local clients including a jewelry shop, hardware store, restaurant, and an indoor playground booking platform
- Delivered end-to-end solutions covering UI design, branding, and front-end development
- Technologies used: React, HTML/CSS, JavaScript

PROJECTS

AVR Autonomous Line-Following Car with Smart Parking (Arduino Uno)

2026

- Programmed an autonomous line-following robot using AVR (ATmega) in Microchip Studio with 3 IR sensors for line tracking and cross-line detection logic
- Implemented a parking bay detection system using an IR receiver — upon single cross-line detection, the car enters parking search mode, detects the bay, and parks autonomously
- On double cross-line detection, the car exits parking mode and resumes normal line following, demonstrating a full FSM-based behavioural control system
- Stack: AVR (ATmega), Arduino Uno, Microchip Studio, Embedded C, IR sensors

PID-Based DC Motor Speed Control System (Raspberry Pi + Python) - [Link](#)

2026

- Designed a real-time closed-loop control system using PID algorithm with encoder feedback for precise RPM measurement and regulation
- Extended the project with a Python desktop GUI application to visualize live motor behaviour, RPM graphs, and PID tuning parameters in real time
- Achieved stable, accurate motor control — successfully demonstrated and documented system performance
- Stack: Python, Raspberry Pi, PID control, encoder interfacing, data visualisation

ESP32-CAM Remote-Controlled Car with Live Video Streaming

2026

Live demo: [Web controller on Vercel](#)

- Built a WiFi-controlled RC car with live video streaming using the ESP32-CAM module as both the access point and video server
- Developed a mobile app and a web-based controller (live on Vercel) allowing real-time driving and camera feed viewing from any device on the network
- Integrated live MJPEG video stream with directional controls, synchronising car movement and video feed over ESP32's internal WiFi
- Stack: ESP32-CAM, Embedded C, JavaScript, Vercel, WiFi/HTTP streaming

IoT Smart Surveillance & Access Control System (ESP32)

2024

- Built a smart surveillance system integrating camera module with Telegram bot for real-time intrusion detection and remote door lock/unlock via commands
- Implemented UART/I2C communication protocols and real-time IoT messaging for reliable remote control
- Stack: ESP32, C/Embedded C, Telegram Bot API, IoT communication

FSM-Based Sequence Detector

2026

- Designed a pseudo-random sequence generator using XOR feedback logic and PISO shift register serialisation
- Implemented a Finite State Machine (FSM) for pattern detection with synchronised clocking

Learning Management System (Spring Boot)

2024

- Developed RESTful backend APIs for authentication and course management with scalable database integration

TECHNICAL SKILLS

Languages: C, Embedded C, AVR Assembly, Python, JavaScript, TypeScript

Frontend: React, Vue.js, Electron.js, HTML/CSS

Backend & DB: Spring Boot (Java), Node.js, PostgreSQL, Docker (basic)

Embedded & Hardware: AVR (ATmega), Arduino, ESP32, Raspberry Pi, UART / SPI / I2C protocols, Oscilloscope & lab instruments, Digital Logic Design, Sensors & Actuators

Tools: Git, Arduino IDE, Proteus, MATLAB, VS Code

CERTIFICATIONS & LEARNING

- The Complete Full-Stack Web Developer Bootcamp — Dr. Angela Yu, Udemy (HTML, CSS, JS, Node.js, React, PostgreSQL)

INTEREST AREAS

Embedded Systems | IoT | Robotics & Control Systems | Full-Stack Development | Industrial Electronics